07/10/26

Tarea 6. Transformada de Laplace de funciones simples.

Determinar $F(s)$ y su RoC cuando:

$a)\ f(t) = t$

$b)\ f(t) = e^{kt}$

$c)\ f(t) = sen(at)$

$d)\ f(t) = \cos(at)$

$Donde\ a \in R, \qquad k \in C$

a) $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt$

$F(s) = \int_0^\infty t\,e^{-st}\,dt = \lim_{b\to\infty} \int_0^b t\,e^{-st}\,dt =$

$u = t \quad dv = e^{-st}\,dt$

$du = dt \quad v = -\frac{1}{s}e^{-st}$

$\lim_{b\to\infty}\left[-\frac{1}{s}te^{-st} + \int_0^b \frac{1}{s}e^{-st}\,dt\right]$

$= \lim_{b\to\infty}\left[-\frac{1}{s}te^{-st} - \frac{1}{s^2}e^{-st}\Big|_0^b\right]$

$= \lim_{b\to\infty}\left[-\frac{1}{s}be^{-sb} - \frac{1}{s^2}e^{-sb} - \left(-\frac{1}{s}(0)e^{-s\cdot\infty} - \frac{1}{s^2}e^{-s(0)}\right)\right]$

$= \lim_{b\to\infty}\left[-\frac{1}{s}be^{-sb} - \frac{1}{s^2}e^{-sb} + \frac{1}{s^2}\right] = -\lim_{b\to\infty}\left[\frac{1}{s}be^{-sb}\right] - \lim_{b\to\infty}\left[\frac{1}{s^2}e^{-sb}\right] + \frac{1}{s^2}$

$= -\frac{1}{s}\lim_{b\to\infty}\left[\frac{b}{e^{sb}}\right] - \frac{1}{s^2}\lim_{b\to\infty}\left[e^{-sb}\right] + \frac{1}{s^2}$

$= -\frac{1}{s^2}\lim_{b\to\infty}\left[e^{-sb}\right] - \frac{1}{s^2}\lim_{b\to\infty}\left[e^{-sb}\right] + \frac{1}{s^2}$

$= -\frac{2}{s^2}\lim_{b\to\infty}\left[e^{-sb}\right] + \frac{1}{s^2}$

$= -\frac{2}{s^2}\lim_{b\to\infty}\left[e^{-\sigma b}\left(\cos(\omega b) + j\,sen(\omega b)\right)\right] + \frac{1}{s^2}$

(oscila entre $[0,1]$; $e^{-\sigma b}\to 0$)

↳ Para $\sigma > 0$ ó $Re(s) > 0$

$s = \sigma + j\omega$

$= 0 + \frac{1}{s^2} = \frac{1}{s^2}$

$\mathcal{L}\{t\} = F(s) = \frac{1}{s^2}, \quad Re(s) > 0$ //

b) $e^{kt}, \quad k \in \mathbb{C}$

$F(s) = \int_0^\infty e^{kt}\,e^{-st}\,dt = \lim_{b\to\infty}\int_0^b e^{(k-s)t} = \lim_{b\to\infty}\left[\frac{1}{k-s}e^{(k-s)t}\Big|_0^b\right]$

$= \lim_{b\to\infty}\left[\frac{1}{k-s}e^{(k-s)b} - \frac{1}{k-s}e^{(k-s)(0)}\right] = \lim_{b\to\infty}\frac{1}{k-s}\left[e^{(k-s)t} - \frac{1}{k-s}\right]$

$= \frac{1}{k-s}\lim_{b\to\infty}\left[e^{(k-s)b} - 1\right]$

$= \frac{1}{k-s}\lim_{b\to\infty}\left[e^{(\alpha-\sigma)b}\left(\cos(\beta-w) + j\,sen(\beta-w)\right) - 1\right]$

($e^{(\alpha-\sigma)b}\to 0$; oscila entre $[0,1]$)

↳ Para $\alpha - \sigma < 0 \to \sigma > \alpha$ ó $Re(s) > Re(k)$

$s = \sigma + jw$

$k = \alpha + j\beta$

$= \frac{1}{k-s}[-1] = \frac{1}{s-k}$

$\mathcal{L}\{e^{kt}\} = F(s) = \frac{1}{s-k}, \quad Re(s) > Re(k)$ //

c) $f(t) = sen(at), \quad a \in \mathbb{R}$

$sen(at) = \frac{e^{jat} - e^{-jat}}{2j}$

$\mathcal{L}\left\{\frac{e^{jat} - e^{-jat}}{2j}\right\} = \int_0^\infty \left(\frac{e^{jat} - e^{-jat}}{2j}\right)e^{-st}\,dt$

$= \frac{1}{2j}\lim_{b\to\infty}\int_0^b \left(e^{-(s-ja)t} - e^{-(s+ja)t}\right)dt$

$= \frac{1}{2j}\lim_{b\to\infty}\left[\frac{e^{-(s-ja)t}}{-(s-ja)} - \frac{e^{-(s+ja)t}}{-(s+ja)}\Big|_0^b\right]$

$= \frac{1}{2j}\lim_{b\to\infty}\left[\frac{e^{-(s-ja)b}}{-(s-ja)} - \frac{e^{-(s+ja)b}}{-(s+ja)} - \left(\frac{e^{-(s-ja)(0)}}{-(s-ja)} - \frac{e^{-(s+ja)(0)}}{-(s+ja)}\right)\right]$

$= \frac{1}{2j}\lim_{b\to\infty}\left[\frac{-e^{-(s-ja)b}}{s-ja} + \frac{e^{-\sigma b}\,e^{-(w+ja)b}}{s+ja} - \left(\frac{1}{s-ja} + \frac{1}{s+ja}\right)\right]$

($e^{-(s-ja)b}$ oscila entre $[0,1]$; $e^{-\sigma b}\to 0$)

$s = \sigma + jw$ para $\sigma > 0$ ó $Re(s) > 0$

$= \frac{1}{2j}\left[\frac{-1}{s+ja} + \frac{-1}{s-ja}\right] = \frac{-s+ja-s-ja}{(s+ja)(s-ja)}\left(\frac{1}{2j}\right)$

$= \frac{2ja}{(s+ja)(s-ja)}\left(\frac{1}{2j}\right)$

$= \frac{a}{(s+ja)(s-ja)} = \frac{a}{s+a^2}$

$\mathcal{L}\{sen(at)\} = F(s) = \frac{a}{s^2+a^2}, \quad R(s) > 0$ //

d) $f(t) = \cos(at)$, $a \in \mathbb{R}$

$$\cos(at) = \frac{e^{jat} + e^{-jat}}{2}$$

$$\mathcal{L}\left\{\frac{e^{jat} + e^{-jat}}{2}\right\} = \int_0^{\infty} \left(\frac{e^{jat} + e^{-jat}}{2}\right) e^{-st}\,dt$$

$$= \frac{1}{2} \lim_{b\to\infty} \int_0^b \left(e^{-(s-ja)t} + e^{-(s+ja)t}\right) dt$$

$$= \frac{1}{2} \lim_{b\to\infty} \left[\frac{e^{-(s-ja)t}}{-(s-ja)} + \frac{e^{-(s+ja)t}}{s+ja}\Big|_0^b\right]$$

$$= \frac{1}{2} \lim_{b\to\infty} \left[-\frac{e^{-(s-ja)b}}{s-ja} - \frac{e^{-(s+ja)b}}{s+ja} - \frac{e^{-(s-ja)0}}{s-ja} - \frac{e^{-(s+ja)0}}{s+ja}\right]$$

$$= \frac{1}{2} \lim_{b\to\infty} -\frac{e^{-(s-ja)b}}{s-ja} - \frac{e^{-(s+ja)b}}{s+ja} + \left(\frac{1}{s-ja} + \frac{1}{s+ja}\right)\Big]$$

$$= \frac{1}{2} \lim_{b\to\infty} \left[-\frac{e^{-\sigma b} e^{-(w-ja)b}}{s-ja} - \frac{e^{-\sigma b} e^{-(w+ja)b}}{s+ja} + \left(\frac{1}{s-ja} + \frac{1}{s+ja}\right)\right]$$

$s = \sigma + jw$

$$= \frac{1}{2}\left[\frac{1}{s-ja} + \frac{1}{s+ja}\right] = \frac{1}{2}\left[\frac{s+ja+s-ja}{(s-ja)(s+ja)}\right]$$

$$= \frac{1}{2}\left[\frac{2s}{s^2+a^2}\right] = \frac{s}{s^2+a^2}$$

$$\mathcal{L}\{\cos(at)\} = F(s) = \frac{s}{s^2+a^2},\ \mathrm{Re}(s),\ \mathrm{Re}(s) > 0$$